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Influence of Controller on Cable Driven Lower Limb Rehabilitation Exoskeleton (C-LREX): PD vs MPC*

Rajan Prasad, Kinda Khalaf, Mohammad I. Awad, and Marwan El-Rich

***Abstract* — Cable driven exoskeletons are preferred over link-driven exoskeletons due to numerous advantages such as remote actuation, lightweight, reconfigurability, and many more. PD (proportional derivative) controllers are widely implemented in exoskeletons due to simplicity and faster response. However, in cable driven exoskeleton, the PD controller doesn't directly distribute the cable tension and thus the possibility of tracking error is higher. In this work, we formulated a non-linear model predictive controller (NMPC) for C-LREX that directly allocates cable tension and is further simplified to reduce the time delay. The simulation result indicated that simplified NMPC is faster than NMPC but slower than PD for C-LREX. NMPCs tracked trajectory with the least error, however, allocated higher cable tension. PD should be preferred when speed is prioritized while NMPC should be preferred for performance.**

***Keywords*— cable-driven exoskeleton, rehabilitation, model predictive control, PD control.**

I. INTRODUCTION

Cable-driven rehabilitation devices (CDRDs) are preferred over link-based exoskeletons because they can be remotely controlled, are lightweight, and cause minimal inertia and inertial vibration on the affected limb. Moreover, CDRDs take into account the biomechanics of the joints. Unlike direct actuation-based exoskeletons, CDRDs do not need to be precisely aligned with the joints, which shortens the time needed to put them on and take them off while also increasing safety.

In the majority of lower limb rehabilitation devices, either link-based devices [1]–[4] or cable driven devices [5]–[8], PD/PID-based impedance controllers are implemented to track the desired reference trajectory. Some other controllers, such as force field controllers [5], [9], model predictive controllers [10]–[12], and sliding mode controllers [13], [14] have been implemented in various rehabilitation devices. Usually, the lower limb model is derived considering the *n-link* serial chain where joint moment drives the model. CDRDs generate joint moments by applying cable tension. In PD-controlled CDRD, the controller doesn't acknowledge the cable tension directly during the control allocation, and a separate algorithm is used to distribute the cable tension to meet the joint moments predicted.

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In previous studies [15], [16], a generalized framework was developed for the design and analysis of the Cable-Driven Lower Limb Rehabilitation Exoskeleton (C-LREX) using a 2-degree-of-freedom (DOF) model of the lower limb, specifically focusing on the sagittal plane (hip and knee flexion/extension). This framework was able to incorporate various design parameters such as the number of cables, subject anthropometry, cuff locations, and reference trajectories. Furthermore, a 3 DOF model of the lower limb was developed, including the hip adduction/abduction of the frontal plane [17], and a PD controller was formulated for tracking the reference healthy trajectory. In this study, we constructed an NMPC (nonlinear model predictive control) for the C-LREX and evaluate it against the PD controller and examined how different controllers impact C-LREX's performance (trajectory tracking) and requirement (cable tension distribution).

II. C-LREX 3DOF MODEL

In the previous study [17], it was found that 4 cable-driven model of C-LREX successfully tracked the bi-planar reference healthy trajectory and was selected as a conceptual model for this study. The lower limb is modelled as a 2-link pendulum (Fig 1) and assumed that the foot is fixed perpendicular to the shank. The model is assumed to aid only during the swing phase of the gait. The generalized equation of the dynamics is derived in [17] as:

$$M(q)\ddot{q}+C(q,\dot{q})\dot{q}+G(q)=\tau \tag{1}$$

The variable $q=[\phi_1 \quad \theta_1 \quad \theta_2]^T$ is the generalized coordinate, while the hip adduction angle is denoted by ϕ_1 and the hip and knee flexion angles are indicated by θ_1 and θ_2 respectively.

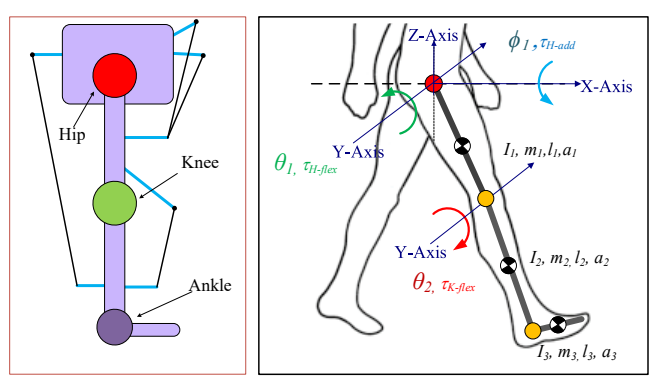

Fig. 1. C-LREX and 3DOF model of the lower limb [17]

In the equation (1), $M \in \mathbb{R}^{3\times3}$, $C \in \mathbb{R}^{3\times1}$, *and* $G \in \mathbb{R}^{3\times1}$ *refer to* the inertial matrix, Coriolis component, and Gravitational

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components respectively. The torques acting at the joints are $\tau = \begin{bmatrix} \tau_{H-add} & \tau_{H-flex} & \tau_{K-flex} \end{bmatrix}^T \in \mathbb{R}^{3\times1}$.

A. Joint torques estimation:

The joint moment in equation (1) is the sum of the moments generated by the applied cable tension and the passive elastic joint moment contributed by the limb.

1) Cable Tension to Joint Torque Mapping

The magnitude of force and routing of the cable are critical factors in determining the joint moment produced. To approximate the Jacobian matrix for each cable, a vector projection technique was employed as detailed in references [15], [16]. These approximations of Jacobian were then utilized to construct a control performance matrix, which evaluates the overall conversion of cable tension to an equivalent joint moment. To determine the joint forces caused by the cable tension, unit vectors were determined by utilizing the limb segment's geometric configuration and joint angles. Subsequently, the cable tension vectors were projected onto these unit vectors to estimate the joint forces and compute the corresponding joint moment utilizing the cross-product of the distance and cable tension vectors. The relationship between joint moment and cable tension is expressed as follows [17]:

$$\tau_{cable} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \begin{bmatrix} J_{11} \\ J_{21} \\ J_{31} \end{bmatrix} F = J^T F \tag{2}$$

Where J_{ij} is Jacobian that maps the cable tension to the joint moment.

Depending on the cuff locations and lower limb joint angles, the control performance matrix B (dependent on the geometric configuration and the number of cables, and keep changing throughout the gait cycle) for 4 cables can be found as:

$$F = \begin{bmatrix} F_1 & F_2 & F_3 & F_4 \end{bmatrix}^T, \tau_{cable} = \begin{bmatrix} \tau_1 & \tau_2 & \tau_3 \end{bmatrix}^T = B^T F$$

$$B = \begin{bmatrix} J_1 & J_2 & J_3 & J_4 \end{bmatrix}^T = \begin{bmatrix} J_{11} & J_{12} & J_{13} & J_{14} \\ J_{21} & J_{22} & J_{23} & J_{24} \\ J_{31} & J_{32} & J_{33} & J_{34} \end{bmatrix}^T \tag{3}$$

$$B \in \mathbb{R}^{4\times3}, J \in \mathbb{R}^{3\times1}$$

B. Passive Joint Moment Estimation

The passive elastic joint moment [18], which is produced by the joint's surrounding tissues, is considered the user's passive contribution in the sagittal plane, while no such contribution is assumed in the frontal plane. The passive elastic joint moment is estimated using a double exponent model developed by Riener et al. [18] which is dependent on the joint angle at the instant of the motion.

Thus, the total joint moment can be written as:

$$\tau = \tau_{cable} + \tau_{passive} = B^T F + \tau_{passive} \tag{4}$$

III. Controller Formulation

PD and NMPC controllers are formulated to track the trajectory with C-LREX. PD controller assigns control based on the error during tracking while NMPC employs an optimization approach to minimize the error along with control being allocated.

A. PD Controller

The study employed a PD-based impedance controller to estimate the desired torque by considering joint errors, as in previous studies [1]–[3]. The controller, illustrated in Fig. 2, generated the required torque for following the subsequent step based on the current step during tracking. To calculate the joint moment, various inputs were utilized, including the current angles, angular velocities, inertial and other matrices (M, C, G), as well as the user's voluntary input based on C-LREX output data. The desired torque was then allocated among the cables as cable tension, considering the system's ability and constraints during that motion instant. The error dynamics system was transformed into an exponentially stable system by utilizing PD-based feedback linearization as per the following expression:

$$\tau = M\left(K_p e + K_d \dot{e} + \ddot{q}_d\right) + C\dot{q} + G \tag{5}$$

Where K_p and K_d are diagonal matrices.

$$K_p = diag\left(K_{p1}, K_{p2}, K_{p3}\right), K_d = diag\left(K_{d1}, K_{d2}, K_{d3}\right)$$

The error dynamics are found as:

$$K_p e + K_d \dot{e} + \ddot{e} = 0 \tag{6}$$

The joint moment estimated by the PD controller is fulfilled in two parts: cable-induced joint moment and passive elastic joint moment. By removing the passive elastic joint moment at the instant of the motion, the remaining joint moment (τ-$\tau_{passive}$) is expected to be delivered using the cable tensions. Moreover, the joint moments that cables can generate are dependent on the geometric configurations and applied cable tension. Thus, the cable tension is to be distributed optimally so that the cable can generate moments equal to or close to expected.

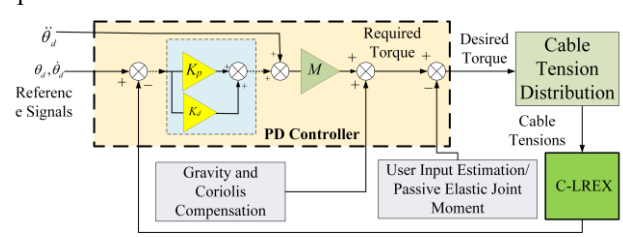

Fig. 2. Schematic layout of the PD control

1) Cable Tension Distribution

The dimension of the control performance matrix B can either be square or rectangular, depending on the number of cables utilized. Consequently, determining cable tension based on the desired joint moment may not be a straightforward task. To tackle this issue, the problem of cable tension allocation is formulated as a hybrid optimization problem, with the primary objective being the minimization of error, while the secondary objective is to minimize control effort. By solving this hybrid optimization problem, a solution is guaranteed to exist in all possible scenarios, especially when there is no unique solution in the primary objective [19].

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$$\min_F J(F) = \left\| W_v \left(B^T F - \left(\tau - \tau_{passive} \right) \right) \right\| + \gamma \left\| W_F F \right\| \quad (7)$$
$$s.t. \quad F_{\min} \le F \le F_{\max}$$

Where W_v and W_F are weight matrices, and γ is the weight ($0< \gamma<1$) that combines primary and secondary optimization problems.

To solve the hybrid optimization problem described above, it is transformed into a standard QP function represented by equation (7). The solution to this problem can be obtained using either the MATLAB QP solver or the algorithm outlined in [19].

$$\min_F \left(\frac{1}{2} F^T H F + f^T F \right) \quad (8)$$
$$s.t. \{ AF \le D$$

Where $A = \begin{bmatrix} -I & I \end{bmatrix}^T, D = \begin{bmatrix} -F_{\min} & F_{\max} \end{bmatrix}^T$

The gradient and Hessian matrix (***f*** and ***H***) are derived from the hybrid objective function.

B. Non-linear model predictive controller (NMPC):

A non-linear model predictive controller is formulated with an objective function that minimizes error in trajectory tracking and the control allocated. A schematic diagram of NMPC is shown in Fig. 3. Unlike PD, the cable tension is distributed implicitly by MPC without requiring an additional cable tension distribution module.

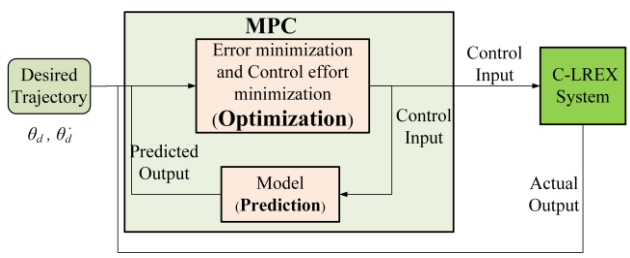

Fig. 3. General layout of MPC control

The objective function for NMPC can be written as:

$$\min_F \quad J = \frac{1}{2} \sum_{k=1}^{N_p} (r_k - q_k)^T Q (r_k - q_k) + \frac{1}{2} \sum_{k=1}^{N_c} F_{k-1}^T P F_{k-1} \quad (9)$$
$$s.t. \quad F_{\min} \le F \le F_{\max}$$

Where, N_p, N_c is the prediction and control horizon, F is the cable tension, r_k, and q_k are the reference and predicted trajectory at k^{th} timestep, P and Q are weight matrices.

The predicted trajectory is calculated via the dynamic equation of motion as:

$$\begin{aligned} q_{k+1} &= \left(M(q_k) \right)^{-1} \left(-C(q_k, \dot{q}_k) \dot{q}_k - G(q_k) + \tau_k \right) \\ &= \left(M(q_k) \right)^{-1} \left(-C(q_k, \dot{q}_k) \dot{q}_k - G(q_k) + \tau_k^{passive} + B_k^T F_k \right) \end{aligned} \quad (10)$$

Where, for the receding horizon problem, the input F is unique only for the control horizon (N_c). The predicted output of the model is then used to determine the optimal control input to apply to the system. To estimate output beyond the control horizon (N_p>N_c), the constant input at the end of the control horizon (N_c) is considered for remaining.

IV. Simulation Result and Discussion

The lower limb model and both controllers were implemented in MATLAB with the listed parameters in Table I. Matrix estimations are simplified in NMPC to reduce time to solve the optimization problem.

Table I. Controller Parameters

Type	Parameters Name	Values
PD Gains	K_p	$diag(300, 240, 200)$
	K_d	$diag(40, 30, 20)$
Gait	Gait Time	3.48 *seconds*
User	Mass	78.5 *kg*
	Height	179.3 *cm*
NMPC parameters	Q (for each set of predictions)	$100 \times diag(100, 10, 1, 10, 1, 1)$
	P (for each set of control)	$0.0000001 \times eye(nu)$
	N_p *(Prediction horizon)*	3
	N_c *(control horizon)*	2
	Optimization Solver	SQP (sequential quadratic programming)
	Maximum Iterations	50000
	Tolerance	1e-6

A. Assumptions and Simplifications in NMPC Controller:

NMPC is a type of open-loop controller that solves an optimization problem at each time step. During optimization, matrices such as *M, C, G, B,* and passive joint moments, which are dependent on the system's states, are estimated repeatedly. This results in a higher time to solve the optimization. To reduce this delay, these matrices are estimated only for each time step and assumed to be constant during each iteration's prediction horizons. However, because these matrices vary at each iteration's time step, the MPC problem remains nonlinear, and optimization is convergence to a feasible solution during each timestep. The sampling time is assumed to be *0.01 seconds*, and the prediction and control horizons are small. During the prediction horizon, small changes in the trajectory angle can be ignored, and consequently, the changes in *M, G, B, and* $\tau_{passive}$ can be disregarded.

B. Tracking Result

The reference trajectory (shown in Fig. 4) is adapted from the dataset published by Fukuchi et al [20].

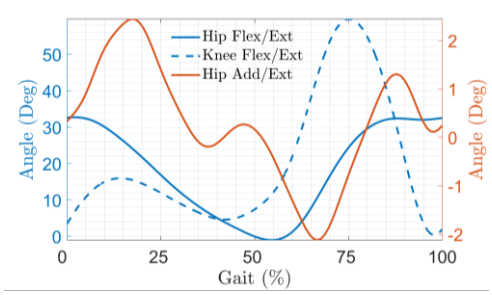

Fig. 4. Bi-planar reference trajectory

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The detailed anthropometric data is estimated using Winter [21] model based on user height and mass. Fig. 5 shows trajectory tracking results using PD and NMPC controllers for C-LREX. NMPC and simplified NMPC have similar tracking performance, but PD has a larger error due to separate cable tension distribution, unlike NMPC. However, PD-based C-LREX required lesser time than MPC for both overall gait tracking and solving optimization problem at each timestep. PD-controlled C-LREX successfully solved the optimization problem within a sampling time while simplified NMPC takes almost 14 sampling times and NMPC performed the slowest (Table II). PD estimates joint moments and distributes them as cable tension using QP programming, resulting in a larger error in tracking if the cable system cannot generate the desired moment depending on the control performance matrix of the configuration at the instant of motion. This limits the capability of PD controlled C-LREX model to meet the predicted desired moment closely as shown in Fig. 6. Thus, employing PD or PID or similar controller type (SMC) where the joint moment is generated as controller output will suffer from the limitation and will result in higher tracking error. If the joint moment estimated by the PD controller is to be applied directly at the joint instead of employing cable, the model tracks the reference trajectory similar to the NMPC case with the least error.

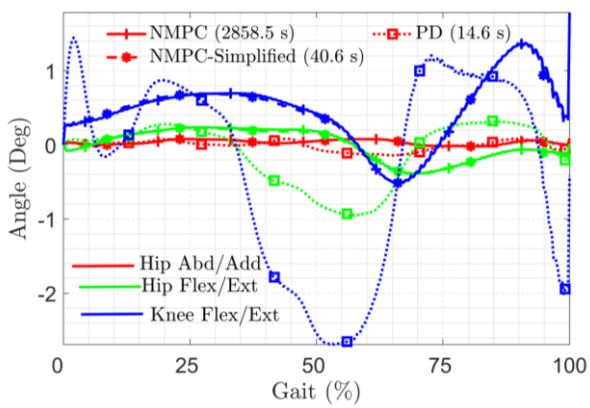

Fig. 5. Trajectory tracking error with different controllers

Table II. Optimization Time Elapsed in one Sample Time

Controller	NMPC	NMPC Simplified	PD
Time (s)	10.8	0.14	<0.01

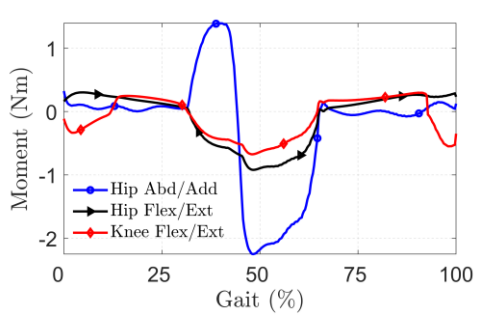

Fig. 6. Error in desired joint moment and cable generated joint moment in PD controller

The cable tension required in different controllers is shown in Fig. 7. PD controlled model tends to allocate a smooth pattern of cable tension compared to others due to a smooth pattern in the desired joint moment. NMPC controller's cable tension pattern tends to be noisy since the cable tension is a variable in the main optimization function and includes the control performance matrix implicitly. PD-controlled C-LREX tends to allocate a slightly lower magnitude of cable tension compared to NMPC controllers and thus a relatively larger error is observed in the trajectory tracking (due to the difference in the desired and supplied moment by cable tension). Nevertheless, it is worth noting that the cable tension pattern in both NMPC controllers is similar with a huge execution time difference.

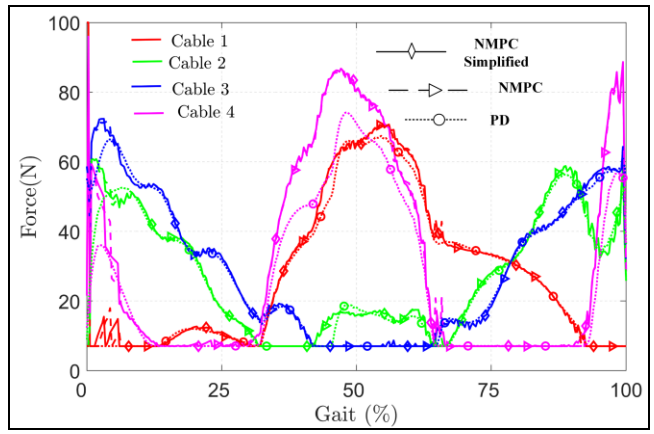

Fig. 7. Cable tensions required in different controllers

1) Alteration in reference trajectory and Anthropometric data:

The reference trajectory in Fukuchi dataset [20] contains respective subject anthropometric information. In this case, we choose a different reference trajectory and respective anthropometric measurements as listed in Table III. Controller parameters remain unchanged. The simplified NMPC controller is used instead of the NMPC controller due to a relatively faster response. The PD controller remains unchanged.

Table III. Changes in Trajectory and Anthropometric Data

Parameters Type	Name	Values
Trajectory	Gait Time	3.60 *seconds*
User's Anthropometry	Mass	61.15 *kg*
	Height	170.50 *cm*

The joint moment required to drive the lower limb model depends on the trajectory (magnitude), trajectory speed (gait time), and user anthropometric data (segments mass, length, and inertia). Furthermore, the moment that the cable system can generate depends on the control performance matrix (geometric configuration) and applied cable tension. Some reference trajectories may require higher joint moments and thus the cable tension limit is to be increased to track the trajectory.

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Tracking performance with user anthropometric data and gait time mentioned in TABLE III is depicted in Fig. 8.

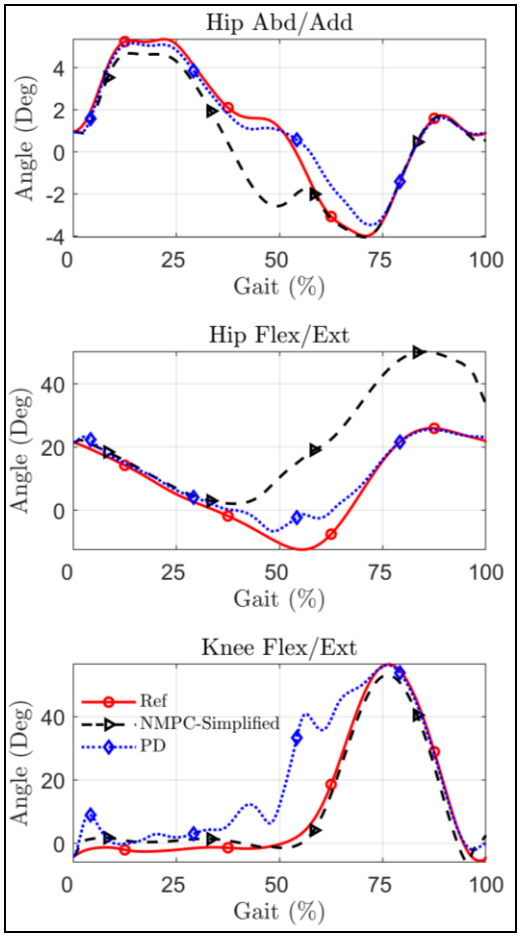

Fig. 8. Trajectory tracking with 100N cable tension

The cable tension was limited to 100N, and the model fails to track the trajectory. The failure to track the trajectory is due to a huge difference between desired and supplied cable based joint moments as depicted in Fig. 9. NMPC also failed to track the trajectory due to insufficient cable tension.

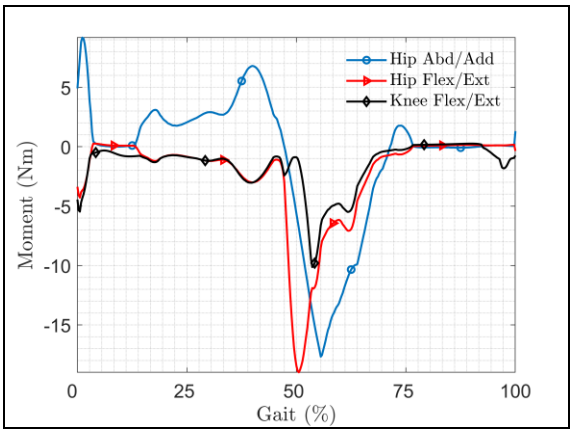

Fig. 9. Error in the desired joint moment and cable generated joint moment in PD controller

The cable limit is increased to 200N, the model is simulated, and trajectory tracking, and cable tension are shown in Fig. 10 and Fig. 11.

The tracking error in PD is similar to the phenomenon observed earlier where PD applies a smaller joint moment than desired via cable and thus the maximum cable tension in the PD controller is less than the NMPC controller. Despite slower response compared to PD, NMPC tends to perform better in trajectory tracking.

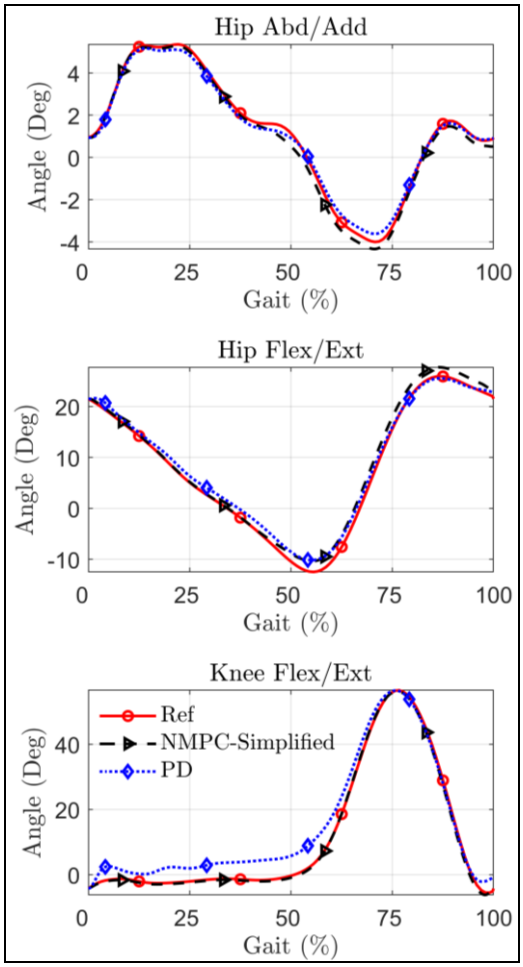

Fig. 10. Trajectory tracking with 200N cable tension (execution time: PD-14 s, NMPC-42 s)

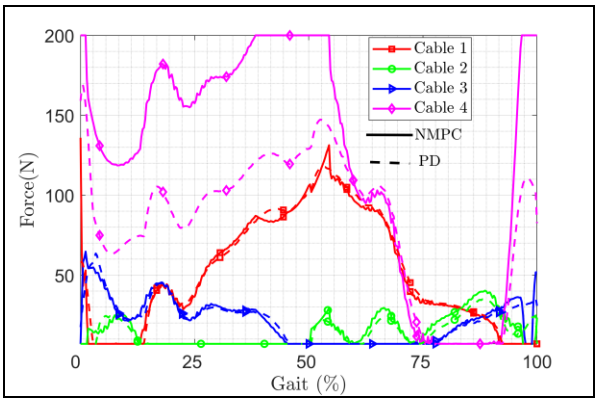

Fig. 11. Cable tension distribution in different controllers with 200N bound

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V. Conclusion

In the majority of exoskeletons, a PD controller is employed to track the trajectory. In the case of cable driven exoskeleton, an additional optimization-based approach cable tension distribution algorithm is required to distribute the cable tension along with the PD controller. Depending upon the geometric configurations, the moment required in C- LREX and the moment generated by cable may not be the same and may lead to errors in trajectory tracking. In this work, we derived a non-linear model predictive controller for C-LREX and is further simplified to reduce the execution time delay. Unlike the PD controller, the NMPC controller allocates cable tension directly instead of moments and then cable tension. Though NMPC outperforms the PD controller in tracking performance, it tends to be slower than the PD controller for C-LREX. Since PD-driven C-LREX primarily focused on joint moment instead of cable tension (unlike in NMPC driven model), the error in tracking is higher compared to NMPC driven. NMPC is preferred over PD when performance is prioritized, while PD is preferred when speed and computational resources are prioritized.

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